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INSTRUCTOR

Homework 4 (4/5) (Homework)

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Blinn College

Due Date: THU, APR 9, 2026 4:15 PM CDT

REQUEST EXTENSION

Current Score: 5 / 51 POINTS | 9.8 %

Scoring and Assignment Information



Calculate the following.

- (a) q when a system does 47 J of work and its energy decreases by 71 J

   J Correct!

- (b) ΔE for a gas that releases 35 J of heat and has 117 J of work done on it

   J Correct!

Solution or Explanation

Because heat and work are the only possible forms of energy transfer, we can attribute the overall change in energy, ΔE , of a system to these two components.

Heat is commonly designated as q and work as w , so we can write $\Delta E = q + w$.

Energy transferred into a system is given a positive sign and energy flowing out of a system carries a negative sign.

Thus when heat flows into a system from the surroundings, the value of q is positive, and when work is done on a system, the value of w is positive.

Conversely, when heat flows out of a system or work is done by the system on the surroundings, q and w will be negative.

- (a) q when a system does 47 J of work and its energy decreases by 71 J

$$\begin{aligned} q &= \Delta E - w \\ &= (-71 \text{ J}) - (-47 \text{ J}) \\ &= -24 \text{ J} \cong -24 \text{ J} \end{aligned}$$

- (b) ΔE for a gas that releases 35 J of heat and has 117 J of work done on it

$$\begin{aligned} \Delta E &= q + w \\ &= (-35 \text{ J}) + (117 \text{ J}) \\ &= 82 \text{ J} \cong 82 \text{ J} \end{aligned}$$

Resources

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[Periodic Table](#)

In which case is heat added to the system?

- ☐ $\Delta E = -79 \text{ J}, w = 97 \text{ J}$
- ☒  $\Delta E = 35 \text{ J}, w = -54 \text{ J}$
- ☐ Heat is added in both of the above cases.
- ☐ Heat is added in neither of the above cases.



✓ Correct!

Solution or Explanation

Because heat and work are the only possible forms of energy transfer, we can attribute the overall change in energy, ΔE , of a system to these two components.

Heat is commonly designated as q and work as w , so we can write $\Delta E = q + w$.

Energy transferred into a system is given a positive sign and energy flowing out of a system carries a negative sign.

Thus when heat flows into a system from the surroundings, the value of q is positive, and when work is done on a system, the value of w is positive.

Conversely, when heat flows out of a system or work is done by the system on the surroundings, q and w will be negative.

$$(1) \quad \Delta E = -79 \text{ J}, w = 97 \text{ J}$$

$$\begin{aligned} q_1 &= \Delta E_1 - w_1 \\ &= (-79 \text{ J}) - (97 \text{ J}) \\ &= -176 \text{ J} \end{aligned}$$

The sign for q is negative, so heat is lost by system 1.

$$(2) \quad \Delta E = 35 \text{ J}, w = -54 \text{ J}$$

$$\begin{aligned} q_2 &= \Delta E_2 - w_2 \\ &= (35 \text{ J}) - (-54 \text{ J}) \\ &= +89 \text{ J} \end{aligned}$$

The sign for q is positive, so heat is added to system 2.

Heat is only added to the second system.

Resources[Read It](#)[Periodic Table](#)

BHChemEng5 9.4.EOC.031.

A metal radiator is made from 31.0 kg of iron. The specific heat of iron is $0.449 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1}$.

How much heat (in J) must be supplied to the radiator to raise its temperature from 21.0°C to 77.0°C ?

4.0 7.79e5 7.79e+05 J Correct!

Solution or Explanation

Heat flow depends on the heat capacity of the material being heated, the mass of the sample, and the change in temperature.

$$\begin{aligned} q &= m \cdot c \cdot \Delta T \\ &= (31.0 \text{ kg}) \left(\frac{1000 \text{ g}}{1 \text{ kg}} \right) \left(\frac{0.449 \text{ J}}{\text{g} \cdot ^\circ\text{C}} \right) (77.0^\circ\text{C} - 21.0^\circ\text{C}) \\ &= (31.0 \text{ kg}) \left(\frac{1000 \text{ g}}{1 \text{ kg}} \right) \left(\frac{0.449 \text{ J}}{\text{g} \cdot ^\circ\text{C}} \right) (56.0^\circ\text{C}) \\ &= 7.7946 \times 10^{+05} \text{ J} \cong 7.79 \times 10^5 \text{ J or } 7.79\text{e5 J} \end{aligned}$$

Resources[Read It](#)[Periodic Table](#)

BHChemEng5 9.4.EOC.032.

The material typically used to heat metal radiators is water. 2.80×10^5 J of heat must be supplied to the radiator in order to raise its temperature from room temperature to 55.0°C , its operating temperature. (Water has a specific heat of $4.184 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1}$.)

If a boiler generates water at 78.0°C , what mass (in kg) of water is required to heat the radiator?

   kg

Solution or Explanation

The final temperature of the water is the operating temperature of the radiator (heat is transferred from the higher temperature water in the boiler to the lower temperature radiator), so the temperature change for water is:

$$\begin{aligned}\Delta T &= T_f - T_i \\ &= 78.0^\circ\text{C} - 55.0^\circ\text{C} \\ &= 23.0^\circ\text{C}\end{aligned}$$

Now calculate the mass of water.

$$\begin{aligned}q &= m \cdot c \cdot \Delta T \\ m &= \frac{q}{c \cdot \Delta T} \\ &= \frac{2.80 \times 10^5 \text{ J}}{(4.184 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1})(23.0^\circ\text{C})} \\ &= (2.9096 \times 10^3 \text{ g}) \left(\frac{1 \text{ kg}}{1000 \text{ g}} \right) \\ &= 2.9096 \text{ kg} \cong 2.91 \text{ kg}\end{aligned}$$

Resources

[Read It](#)[Periodic Table](#)

A silver nail, a tin nail, and a lead nail of the same mass and initially at the same room temperature are all put into a vessel containing boiling water.

Substance	Specific Heat ($\text{J}\cdot\text{g}^{-1}\cdot^{\circ}\text{C}^{-1}$)
Pb	0.160
Sn	0.210
Ag	0.240

Which one would you expect to reach 100°C first?

- ☐ the silver nail
- ☐ the tin nail
- ☒  the lead nail



✓ Correct!

Solution or Explanation

The specific heat capacity for Pb is lower than that for Sn and Ag.

All else equal, the substance with the lowest heat capacity requires the least amount of energy to heat up. It is therefore expected that the lead nail will gain temperature more quickly and reach 100°C first.

Resources

[Read It](#)

[Periodic Table](#)

6. [- / 1 Points]

0/10 Submissions Used

DETAILS

MY NOTES

ASK YOUR TEACHER

BHChemEng5 9.4.EOC.038.

The energy densities of various types of coal are listed below.

Substance	Energy Density (kJ/g)
anthracite	35
subbituminous	31
bituminous	28
lignite	26

An unknown sample of one of these coal types is burned in an apparatus with a calorimeter constant of 1.3 kJ/°C. When a 0.360 g sample is used, the temperature change is 9.68°C.

Which type of coal is the sample?

- ☐ anthracite
- ☐ subbituminous
- ☐ bituminous
- ☐ lignite

Resources

[Read It](#)[Periodic Table](#)

BHChemEng5 9.5.EOC.044.

Classify the following phase changes as exothermic or endothermic.

(a) freezing

---Select--- ▼

(b) boiling

---Select--- ▼

Resources[Read It](#)[Periodic Table](#)

BHChemEng5 9.5.EOC.045.

What happens to the temperature of a material as it undergoes an endothermic phase change? If heat is added, how can the temperature behave in this manner?

During an endothermic phase change, such as ---Select--- ▼ , energy is ---Select--- ▼ by the system. This energy is used to ---Select--- ▼ the particles. Therefore its temperature ---Select--- ▼ .

Resources[Read It](#)[Periodic Table](#)

BHChemEng5 9.5.EOC.047.

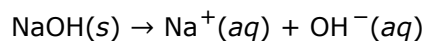
If 16.3 kJ of heat are given off when 2.1 g of HCl condenses from vapor to liquid, what is ΔH_{cond} (in kJ/mol) for this substance?

4.0 ✓

 kJ/mol**Resources**[Read It](#)[Periodic Table](#)

BHChemEng5 9.5.EOC.050.

When a 12.5 g sample of NaOH(s) dissolves in 367.0 mL water in a coffee cup calorimeter, the temperature of the water changes from 21.1°C to 29.9°C.



Assuming that the specific heat capacity of the solution is the same as for water ($4.184 \text{ J} \cdot \text{g}^{-1} \cdot ^\circ\text{C}^{-1}$) and the density of the solution is 1.030 g/mL, calculate the following.

- (a) q_{soln} , the heat transferred from the system to its surroundings (in J)

4.0 ✓ J

- (b) ΔH for the reaction (in kJ)

4.0 ✓ kJ

Resources

[Read It Watch It](#)

[Periodic Table](#)

Write the formation reaction for 1 mol of each of the following substances. (Include states-of-matter under the given conditions in your answer. Use the lowest possible whole number or reduced fractional coefficients.)

(a) $\text{C}_2\text{H}_4(g)$

(b) $\text{C}_3\text{H}_8(l)$

(c) $\text{HCl}(g)$

(d) $\text{C}_5\text{H}_{10}\text{O}_4(s)$

(e) $\text{KF}(s)$

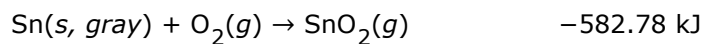
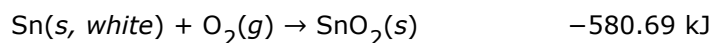
Resources

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BHChemEng5 9.6.EOC.054.

The phase change between white tin and gray tin is difficult to observe directly. Both substances can be burned, however.



From these equations, calculate ΔH° (in kJ) for the conversion of gray tin into white tin.

4.0 ✓

 kJ

Resources

[Read It](#)[Periodic Table](#)

13. [- / 1 Points] MASTERY QUESTION SET

ASK YOUR TEACHER

Get 1 of 2 questions correct in set (Set attempt 1 of 10)

1 ☐2 ☐

BHChemEng5 9.6.EOC.057.

Use the following tabulated thermochemical data to solve this problem.

Substance	ΔH_f° (kJ/mol)	Substance	ΔH_f° (kJ/mol)
$C_5H_{12}(l)$	-173.5	$CO_2(g)$	-393.5
$C_7H_{16}(l)$	-224.2	$H_2O(l)$	-285.8
$C_9H_{20}(l)$	-274.7	$O_2(g)$	0
$C_{11}H_{24}(l)$	-327.2		

The heat of combustion of nonane is -6124.8 kJ/mol. Find the heat of formation of nonane in kJ/mol. (Round your answer to one decimal place.)

 kJ/mol**Resources**[Read It](#)[Periodic Table](#)

SUBMIT ANSWER

14. [- / 1 Points] MASTERY QUESTION SET

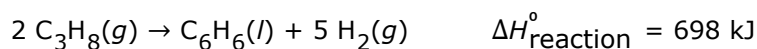
ASK YOUR TEACHER

Get 2 of 3 questions correct in set (Set attempt 1 of 10)

1 ☐2 ☐3 ☐

BHChemEng5 9.7.EOC.063.

Reactions of hydrocarbons are often studied in the petroleum industry. One such reaction is shown below.



If 51 L of propane at 25°C and 0.94 atm is to be reacted, how much heat (in kJ) must be supplied?

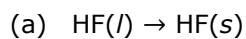
4.0 ✓

 kJ**Resources**[Read It](#)[Periodic Table](#)

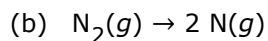
SUBMIT ANSWER

BHChemEng5 9.CP.EOC.073.

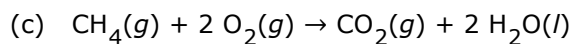
Without looking up any numerical data or doing calculations, predict whether the *enthalpy change* for each of the following reactions should be positive, negative, or zero.



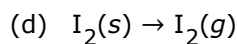
---Select--- ▼



---Select--- ▼



---Select--- ▼



---Select--- ▼

Resources[Read It](#)[Periodic Table](#)

BHChemEng5 9.CP.EOC.078.

When a chemical bond forms, is energy absorbed or released?

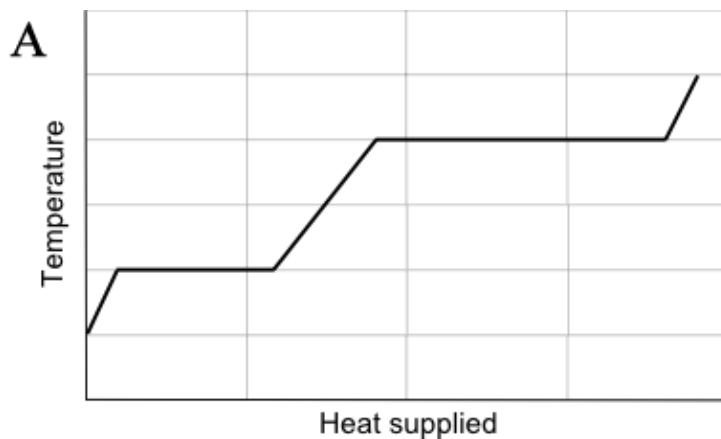
- ☐ absorbed
- ☐ released

Resources[Read It](#)[Periodic Table](#)

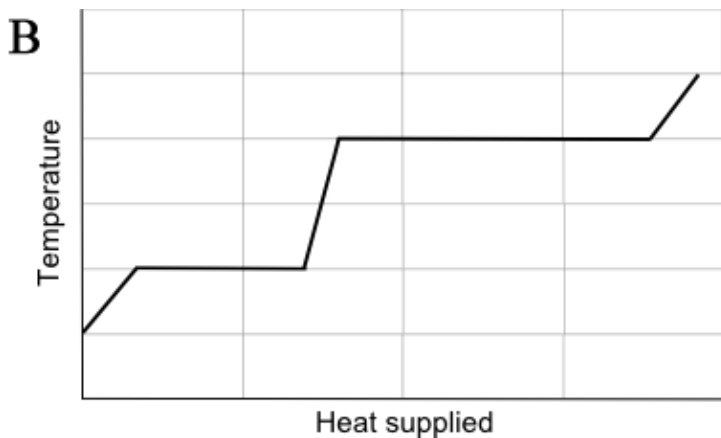
A substance has the following properties

ΔH_{fusion}	$10 \text{ kJ}\cdot\text{mol}^{-1}$
$\Delta H_{\text{vaporization}}$	$20 \text{ kJ}\cdot\text{mol}^{-1}$
$C_p(\text{solid})$	$60 \text{ kJ}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
$C_p(\text{liquid})$	$30 \text{ kJ}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
$C_p(\text{gas})$	$60 \text{ kJ}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$

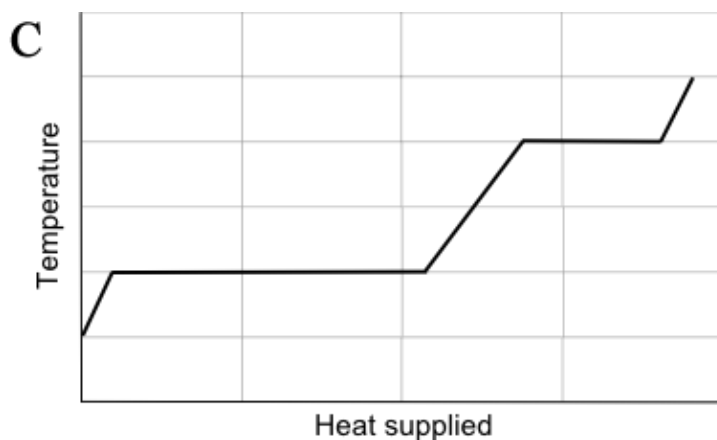
Now examine the following four graphs.



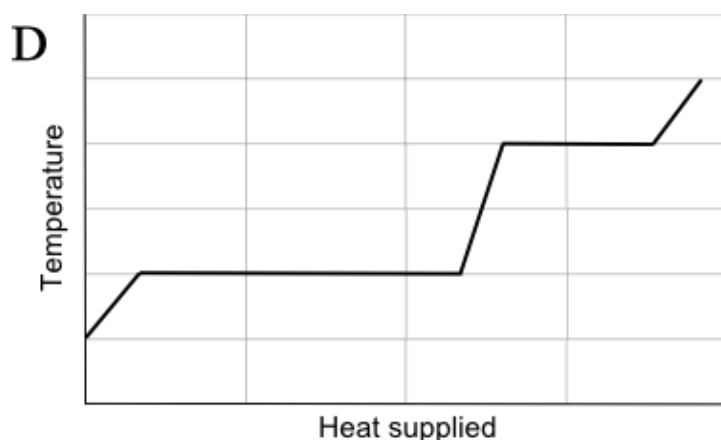
i



i



i



i

Which of the four graphs would be most consistent with these data?

- ☐ Graph A
- ☐ Graph B
- ☐ Graph C
- ☐ Graph D

Explain how you are able to identify the correct graph.

The heat flow from vaporization (second horizontal line) is the heat flow from fusion (first horizontal line). The heat supplied to raise the temperature of the solid (first diagonal line) is the heat supplied to raise the temperature of the gas (third diagonal line). This is also indicated by the values of molar heat capacity for the solid and gas. Additionally, those heat flows are that required to heat the liquid.

Resources

[Read It](#)[Periodic Table](#)

BHChemEng5 9.CP.EOC.083.

A student performing a calorimetry experiment combined 100.0 mL of 0.50 *M* HCl and 100.0 mL of 0.50 *M* NaOH in a coffee cup calorimeter. Both solutions were initially at 20.0°C, but when the two were mixed, the temperature rose to 23.2°C.

- (a) Suppose the experiment is repeated in the same calorimeter, but this time using 200.0 mL of 0.50 *M* HCl and 200.0 mL of 0.50 *M* NaOH.

Will the ΔT observed be greater than, less than, or equal to that in the first experiment, and why?

The reaction of 200.0 mL of 0.50 *M* HCl and 200.0 mL of 0.50 *M* NaOH has amount of reactants distributed over volume, and heat produced compared to the reaction of 100.0 mL of 0.50 *M* HCl and 100.0 mL of 0.50 *M* NaOH. Therefore, the temperature change of the reaction of 200.0 mL acid and 200.0 mL base is the temperature change of the reaction of 100.0 mL acid and 100.0 mL base.

- (b) Suppose the experiment is repeated once again in the same calorimeter, but this time using 100.0 mL of 1.00 *M* HCl and 100.0 mL of 1.00 *M* NaOH.

Will the ΔT observed be greater than, less than, or equal to that in the first experiment, and why?

The reaction of 100.0 mL of 1.00 *M* HCl and 100.0 mL of 1.00 *M* NaOH has amount of reactants distributed over volume, and heat produced compared to the reaction of 100.0 mL of 0.50 *M* HCl and 100.0 mL of 0.50 *M* NaOH. Therefore, the temperature change of the reaction of 1.00 *M* acid and 1.00 *M* base is the temperature change of the reaction of 0.50 *M* acid and 0.50 *M* base.

Resources

[Read It Watch It](#)[Periodic Table](#)

BHChemEng5 10.2.EOC.011.

Identify each of the following processes as spontaneous or nonspontaneous.

- (a) A solution of salt water is poured into a flask of water, and the two mix.
- ☐ This process is spontaneous.
 - ☐ This process is nonspontaneous.
- (b) Nitrogen molecules dissociate to form nitrogen atoms.
- ☐ This process is spontaneous.
 - ☐ This process is nonspontaneous.
- (c) Silver nitrate is added to a solution of potassium bromide and a precipitate forms.
- ☐ This process is spontaneous.
 - ☐ This process is nonspontaneous.
- (d) A tray of water is placed outside during a blizzard and boils.
- ☐ This process is spontaneous.
 - ☐ This process is nonspontaneous.
- (e) The acetic acid in some spilled vinegar evaporates into a kitchen, making the room smell.
- ☐ This process is spontaneous.
 - ☐ This process is nonspontaneous.

Resources

[Read It](#)[Periodic Table](#)

BHChemEng5 10.2.EOC.013.

Which of the following exothermic processes are not spontaneous under some circumstances? (Select all that apply.)

- ☐ Monomers react to form a polymer.
- ☐ Liquid water condenses from water vapor.
- ☐ Snow forms from liquid water.
- ☐ Fossil fuels burn to form carbon dioxide and water.

Resources

[Read It](#)[Periodic Table](#)

BHChemEng5 10.3.EOC.021.

For each pair of items, tell which has the higher entropy and explain why.

- (a) item 1: a sample of O₂ vapor at 0°C
item 2: a sample of O₂ vapor at 20°C

---Select--- ▼ has the higher entropy. Molecules at ---Select--- ▼ temperatures have higher entropies.

- (b) item 1: solid table salt
item 2: the same table salt dissolved in soup

---Select--- ▼ has the higher entropy. The ---Select--- ▼ molecules have ---Select--- ▼ disorder and therefore have higher entropies.

- (c) item 1: a 100 mL sample of pure water and a 100 mL sample of pure alcohol
item 2: the same samples of water and alcohol after they have been poured together and stirred

---Select--- ▼ has the higher entropy. The ---Select--- ▼ molecules have ---Select--- ▼ disorder and therefore have higher entropies.

Resources

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BHChemEng5 10.3.EOC.022.

For each process, tell whether the entropy change of the system is positive or negative.

(a) A glassblower heats glass (the system) to its softening temperature.

- ☐ positive
☐ negative

(b) Dry ice (the system) sublimates into carbon dioxide vapor at room temperature.

- ☐ positive
☐ negative

(c) Spun liquid sugar cools and hardens into a solid cake decoration. (The system consists only of sugar.)

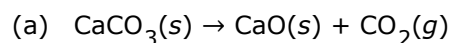
- ☐ positive
☐ negative

Resources

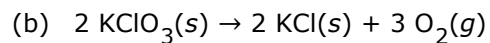
[Read It](#)

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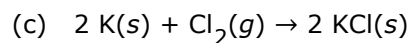
For the following chemical reactions, predict the sign of ΔS for the system. (Note that this should not require any detailed calculations.)



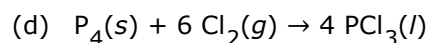
- ☐ positive
☐ negative



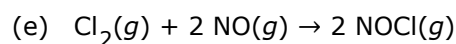
- ☐ positive
☐ negative



- ☐ positive
☐ negative



- ☐ positive
☐ negative



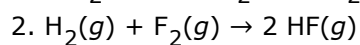
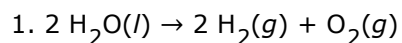
- ☐ positive
☐ negative

Resources

[Read It](#)

[Periodic Table](#)

Which reaction occurs with the greater increase in entropy? Explain your reasoning.



The first reaction has mol of reactants in the gas phase and mol of products in the gas phase.

The second reaction has mol of reactants in the gas phase and mol of products in the gas phase.

Since the number of moles in the gas phase for the first reaction and for the second reaction, the reaction occurs with the greater increase in entropy.

Resources

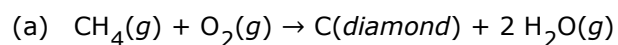
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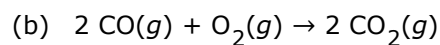
Use the following tabulated thermodynamic data to answer the questions below.

Substance	S° ($\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$)
$\text{C}(\text{diamond})$	2.4
$\text{O}_2(\text{g})$	205.0
$\text{CH}_4(\text{g})$	186.2
$\text{H}_2\text{O}(\text{g})$	188.7
$\text{CO}_2(\text{g})$	213.6
$\text{CO}(\text{g})$	197.6

Calculate the entropy change (in J/K) of each of the reactions listed below. (Assume standard thermodynamic conditions of 1.00 atm and 298.15 K.)



4.0 ✓ J/K



4.0 ✓ J/K

Resources

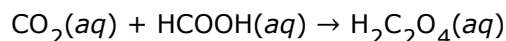
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BHChemEng5 10.5.EOC.039.

Through photosynthesis, plants build molecules of sugar containing several carbon atoms from carbon dioxide. In the process, entropy is decreased.

The reaction of CO_2 with formic acid to form oxalic acid provides a simple example of a reaction in which the number of carbon atoms in a compound increases.



Use the following thermodynamic data to answer the questions below.

Substance	S° ($\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$)
$\text{HCOOH}(aq)$	92.0
$\text{H}_2\text{C}_2\text{O}_4(aq)$	45.61
$\text{CO}_2(aq)$	117.6

- (a) Calculate the standard entropy change (in J/K) for this reaction.

$\Delta S^\circ =$ ☒ J/K

- (b) Discuss the sign of ΔS° .

The entropy change of the reaction is , indicating that molecular disorder is or that the system becomes ordered.

- (c) How do plants carry out reactions that increase the number of carbons in a sugar, given the changes in entropy for reactions like this?

Plants carry out this type of reaction in conjunction with other reactions that entropy. As long as the overall entropy change of the universe is , the coupled reactions can proceed spontaneously.

Resources

[Read It](#)

[Periodic Table](#)

BHChemEng5 10.5.EOC.041.

Rank the following compounds in order of increasing entropy.

- | | | |
|---|---|---|
| ☐ $\text{C}_2\text{H}_2(g)$ | ☐ $\text{C}_2\text{H}_2(g)$ | ☐ $\text{C}_2\text{H}_2(g)$ |
| ☐ $\text{H}_2\text{C}_2\text{O}_4(s)$ | ☐ $\text{H}_2\text{C}_2\text{O}_4(s)$ | ☐ $\text{H}_2\text{C}_2\text{O}_4(s)$ |
| ☐ $\text{C}_6\text{H}_6(l)$ | ☐ $\text{C}_6\text{H}_6(l)$ | ☐ $\text{C}_6\text{H}_6(l)$ |

Explain why this ranking makes sense.

The entropy of a is generally greater than that of a , which in turn is generally greater than that of a .

Resources

[Read It](#)[Periodic Table](#)

BHChemEng5 10.6.EOC.045.

Calculate ΔG° at 45.0°C for reactions for each of the following conditions.

(a) $\Delta H^\circ = 290. \text{ kJ}$; $\Delta S^\circ = -675 \text{ J/K}$

$\Delta G^\circ =$ ☒ kJ

(b) $\Delta H^\circ = -1146 \text{ kJ}$; $\Delta S^\circ = 0.623 \text{ kJ/K}$

$\Delta G^\circ =$ ☒ kJ

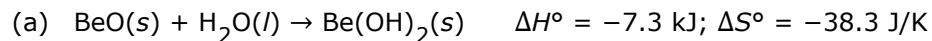
(c) $\Delta H^\circ = -63.8 \text{ kJ}$; $\Delta S^\circ = -441 \text{ J/K}$

$\Delta G^\circ =$ ☒ kJ

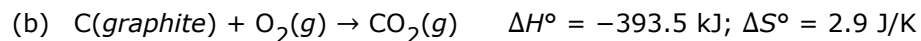
Resources

[Read It](#)[Periodic Table](#)

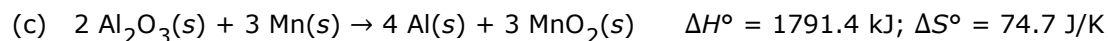
Discuss the effect of temperature change on the spontaneity of the following reactions at 1 atm.



Since ΔH° is and ΔS° is , this reaction is .



Since ΔH° is and ΔS° is , this reaction is .



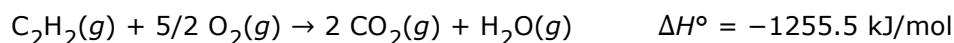
Since ΔH° is and ΔS° is , this reaction is .

Resources

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The combustion of acetylene was used in welder's torches for many years because it produces a very hot flame.



Substance	$S^\circ \left(\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \right)$	Substance	$S^\circ \left(\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \right)$
$\text{C}_2\text{H}_2(g)$	200.8	$\text{H}_2\text{O}(l)$	69.91
$\text{C}_2\text{H}_4(g)$	219.5	$\text{H}_2\text{O}(g)$	188.7
$\text{CO}(g)$	197.6	$\text{O}_2(g)$	205.0
$\text{CO}_2(g)$	213.6	$\text{O}(g)$	161.0
$\text{CO}_2(aq)$	117.6	$\text{O}_3(g)$	238.8

- (a) Use the data in the given table to calculate ΔS° (in J/K) for this reaction.

4.0 ✓ J/K

- (b) Calculate ΔG° (in kJ) and show that the reaction is spontaneous at 25°C.

$\Delta G^\circ = 4.0 \checkmark$ kJ. Since this value is , the reaction is spontaneous at this temperature.

- (c) Is there any temperature range in which this reaction is not spontaneous?

The reaction becomes more spontaneous at temperatures and less spontaneous at temperatures. The reaction is nonspontaneous .

- (d) Do you think you could use the equation, $\Delta G = \Delta H - T\Delta S$, to calculate such a temperature range reliably? Explain your answer.

There is error in using this equation since the ΔH and ΔS terms are for the reaction under standard state conditions and the combustion of acetylene in torches occurs at a temperature the standard state temperature, 25°C.

Resources

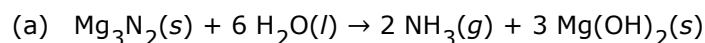
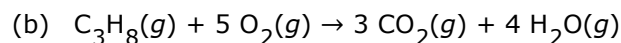
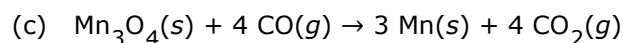
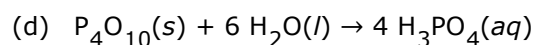
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[Periodic Table](#)

BHChemEng5 10.7.EOC.056.

Using the thermodynamic data below, calculate ΔG° (in kJ) for the following reactions.

Substance	ΔG°_f (kJ·mol ⁻¹)	Substance	ΔG°_f (kJ·mol ⁻¹)	Substance	ΔG°_f (kJ·mol ⁻¹)
C ₃ H ₈ (g)	-23.49	H ₂ O(l)	-237.2	NH ₃ (g)	-16.5
CO(g)	-137.2	H ₂ O(g)	-228.6	O ₂ (g)	0
CO ₂ (g)	-394.4	H ₃ PO ₄ (aq)	-1142.54	P ₄ O ₁₀ (s)	-2698
Mn(s)	0	Mg ₃ N ₂ (s)	-401.1		
Mn ₃ O ₄ (s)	-1283.2	Mg(OH) ₂ (s)	-833.7		

4.0 ✓ kJ4.0 ✓ kJ4.0 ✓ kJ4.0 ✓ kJ**Resources**[Read It](#)[Periodic Table](#)

BHChemEng5 10.7.EOC.059.

The normal melting point of phosphoric acid, H_3PO_4 , is 42.4°C . For the process of melting solid phosphoric acid at 1 atm, what is the sign of each of the following?

(a) ΔH°

- ☐ positive
- ☐ negative
- ☐ zero

(b) ΔS°

- ☐ positive
- ☐ negative
- ☐ zero

(c) ΔG° at 42.4°C

- ☐ positive
- ☐ negative
- ☐ zero

(d) ΔG° at 0.0°C

- ☐ positive
- ☐ negative
- ☐ zero

(e) ΔG° at 50.0°C

- ☐ positive
- ☐ negative
- ☐ zero

Resources

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33. [- / 1 Points] MASTERY QUESTION SET

ASK YOUR TEACHER

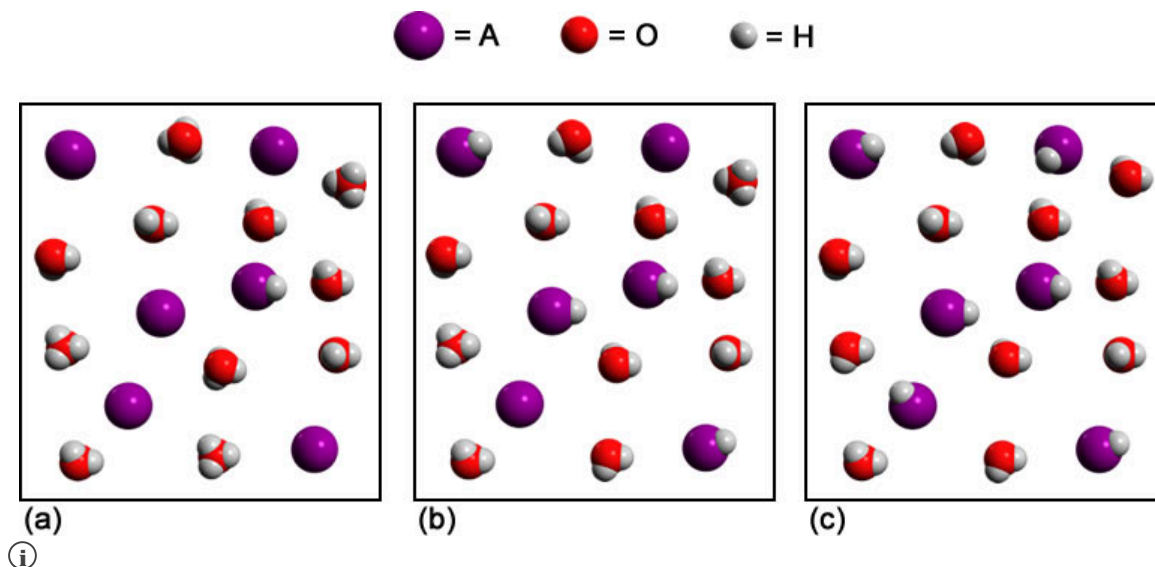
Get 1 of 2 questions correct in set (Set attempt 1 of 10)

1 ☐

2 ☐

BHChemEng5 10.CP.EOC.079.

The molecular scale diagrams below show snapshots of a strong acid, HA, at three different instants after it is added to water.



(a) Place the three panels in the correct order so that they show the progress of the spontaneous process that takes place as the acid dissolves in the water.

- ☐ panel (a), panel (b), panel (c)
- ☐ panel (a), panel (c), panel (b)
- ☐ panel (b), panel (a), panel (c)
- ☐ panel (b), panel (c), panel (a)
- ☐ panel (c), panel (a), panel (b)
- ☐ panel (c), panel (b), panel (a)

(b) Explain your answer in terms of entropy.

What happens to the entropy of this system as the acid dissolves in water?

- ☐ It remains unchanged.
- ☐ It increases.
- ☐ It decreases.

How does entropy change influence whether a process is spontaneous?

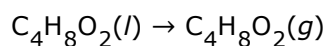
- ☐ A positive entropy change has a negative contribution to the free energy, which must be below zero for a process to be spontaneous.
- ☐ A positive entropy change always results in a process being spontaneous.
- ☐ A negative entropy change always results in a process being spontaneous.
- ☐ A negative entropy change has a negative contribution to the free energy, which must be below zero for a process to be spontaneous.

Resources[Read It](#)[Periodic Table](#)

SUBMIT ANSWER

BHChemEng5 10.A.EOC.082.

Consider the vaporization of ethyl acetate.



Calculate the entropy change, ΔS° (in $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$), for the vaporization of ethyl acetate, $\text{C}_4\text{H}_8\text{O}_2$, at its boiling point of 77.1°C . The heat of vaporization for this compound is 34.1 kJ/mol .

$\Delta S^\circ =$ $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$

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Complete the following statements to make them true.

- (a) An exothermic reaction is usually , but not always. One must also consider changes, which could render an exothermic reaction .
- (b) When ΔG° is positive, the reaction is under the standard conditions, but may be under different conditions.
- (c) ΔS° is positive for a reaction in which there is in the moles of gas.
- (d) If ΔH° and ΔS° are both negative, ΔG° will be at low temperatures and at high temperatures.

Resources

[Read It Watch It](#)

[Periodic Table](#)

The reaction shown below is involved in the refining of manganese.



The table that follows provides all of the thermodynamic data you should need for this problem.

Substance	ΔH°_f ($\text{kJ}\cdot\text{mol}^{-1}$)	S° ($\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$)	ΔG°_f ($\text{kJ}\cdot\text{mol}^{-1}$)
$\text{Mn}_2\text{O}_3(s)$	-959.0	?	-881.5
$\text{C}(s, \text{graphite})$	0	5.740	0
$\text{Mn}(s)$	0	32.0	0
$\text{CO}_2(s)$	-393.5	213.6	-394.4

- (a) Find ΔH° for the reaction (in kJ).

$$\Delta H^\circ = 4.0 \text{ kJ}$$

- (b) ΔS° for the reaction above is 528.97 J/K. Find S° (in $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$) for $\text{Mn}_2\text{O}_3(s)$.

$$S^\circ = 4.0 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$$

- (c) Calculate ΔG° (in kJ) for the reaction at the standard temperature of 298 K. (There are two ways that you could do this.)

using tabular data $\Delta G^\circ = 4.0 \text{ kJ}$

using the Gibbs equation $\Delta G^\circ = 4.0 \text{ kJ}$

- (d) At what temperatures (in K) would this reaction be spontaneous?

$$T \text{ ---Select--- } 4.0 \text{ K}$$

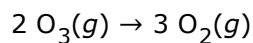
Resources

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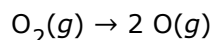
[Periodic Table](#)

BHChemEng5 10.A.EOC.092.

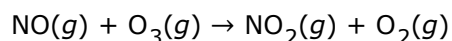
Consider the following set of reactions.



$$\Delta G^\circ = -326 \text{ kJ}$$

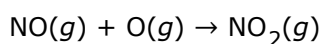


$$\Delta G^\circ = +463.6 \text{ kJ}$$



$$\Delta G^\circ = -198.3 \text{ kJ}$$

Using only the data given above, determine ΔG° (in kJ) for the following reaction.



Remember that ΔG is a state function, just like ΔH .

$$\Delta G^\circ = 4.0 \text{ kJ}$$

Resources

[Read It](#)[Periodic Table](#)

BHChemEng5 11.2.EOC.013.

In each of the following situations, is the rate measured the instantaneous rate or the average rate?

- (a) It took the student 10 minutes to read 5 pages of his textbook, so he read 0.5 pages per minute.

- ☐ average rate
☐ instantaneous rate

- (b) At minute 2.0, the student was reading 0.5 pages of his textbook per minute, but upon reaching a difficult concept at minute 7.5, he slowed down to 0.25 pages per minute.

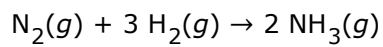
- ☐ average rate
☐ instantaneous rate

Resources

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BHChemEng5 11.2.EOC.016.

The following reaction is for the Haber process, the industrial production of ammonia.



Assume that under certain laboratory conditions ammonia is produced at the rate of $2.62 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$.

At what rate (in $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$) is nitrogen consumed?

4.0 ✓

 $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$

At what rate (in $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$) is hydrogen consumed?

4.0 ✓

 $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$

Resources

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The following data were obtained in the decomposition of $\text{H}_2\text{O}_2(aq)$ to $\text{O}_2(g)$ and $\text{H}_2\text{O}(l)$. (No oxygen was present initially.)

The rate at which oxygen gas was produced was measured.

Time (min)	Volume of O_2 (mL)
3.3	10.7
6.9	18.2

- (a) Calculate the average rate in mL/min for the first 3.3 minutes.

4.0 ✓ mL/min

- (b) Calculate the average rate in mL/min for the first 6.9 minutes.

4.0 ✓ mL/min

Resources

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41. [- / 1 Points] MASTERY QUESTION SET

ASK YOUR TEACHER

Get 1 of 2 questions correct in set (Set attempt 1 of 10)

1 ☐

2 ☐

BHChemEng5 11.3.EOC.029.

The hypothetical reaction, $A + B \rightarrow C$, has the following rate law.

$$\text{rate} = k \cdot [A]^x \cdot [B]^y$$

When $[A]$ is tripled and $[B]$ is held constant, the rate increases ninefold. But the rate triples when $[B]$ is tripled and $[A]$ is held constant. What are the values of x and y ?

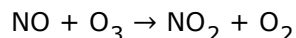
$x =$

$y =$

Resources[Read It Watch It](#)[Periodic Table](#)**SUBMIT ANSWER**

BHChemEng5 11.3.EOC.028.

One reaction that destroys O_3 molecules in the stratosphere is the following.



When this reaction was studied in the laboratory, it was found to be first order with respect to both NO and O_3 , with a rate constant of $1.9 \times 10^4 \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$.

If $[NO] = 1.1 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$ and $[O_3] = 2.5 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$, what is the rate of this reaction?

4.0 ✓

 $\text{L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ **Resources**[Read It](#)[Periodic Table](#)

43. [- / 1 Points] MASTERY QUESTION SET

ASK YOUR TEACHER

Get 2 of 3 questions correct in set (Set attempt 1 of 10)

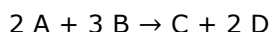
1 ☐

2 ☐

3 ☐

BHChemEng5 11.3.EOC.032.

Consider the following reaction.



The following experimental data were obtained for the reaction.

$[\text{A}] \left(\text{mol} \cdot \text{L}^{-1} \right)$	$[\text{B}] \left(\text{mol} \cdot \text{L}^{-1} \right)$	$\text{rate} = \frac{\Delta[\text{C}]}{\Delta t} \left(\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1} \right)$
0.127	0.15	0.039
0.127	0.30	0.157
0.254	0.15	0.078

Determine the reaction order for each reactant.

reaction order for A

- ☐ first
☐ second
☐ third

reaction order for B

- ☐ first
☐ second
☐ third

Determine the value and the unit of the rate constant.

$k =$

- ☐ $\text{L}^2 \cdot \text{mol}^{-2} \cdot \text{s}^{-1}$
☐ $\text{L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$
☐ $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$
☐ s^{-1}

Resources

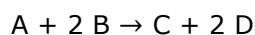
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[SUBMIT ANSWER](#)

BHChemEng5 11.3.EOC.034.

Consider the following generalized reaction.



Rate data were obtained at 25°C from a series of experiments of this reaction.

Expt.	Initial [A] (mol·L ⁻¹)	Initial [B] (mol·L ⁻¹)	Initial Rate of Formation of C (mol·L ⁻¹ ·min ⁻¹)
1	0.136	0.136	4.08×10^{-4}
2	0.404	0.404	1.21×10^{-3}
3	0.136	0.404	4.08×10^{-4}
4	0.275	0.533	8.25×10^{-4}

What is the rate law expression for this reaction? (Rate expressions take the general form: rate = $k \cdot [A]^a \cdot [B]^b$.)

Resources

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45. [- / 1 Points] MASTERY QUESTION SET

[ASK YOUR TEACHER](#)

Get 2 of 3 questions correct in set (Set attempt 1 of 10)

1 ☐2 ☐

3 ○

BHChemEng5 11.4.EOC.044.

A possible reaction for the degradation of the pesticide DDT to a less harmful compound was simulated in the laboratory. The reaction was found to be first-order, with $k = 3.94 \times 10^{-8} \text{ s}^{-1}$ at 25°C .

What is the half-life for the degradation of DDT in this experiment, in years? (On average, there are 365.242 days in a year.)

4.0 ✓

 yr**Resources**[Read It](#)[Periodic Table](#)

SUBMIT ANSWER

BHChemEng5 11.4.EOC.045.

The initial concentration of the reactant in a first-order reaction $A \rightarrow \text{products}$ is 0.449 M and the half-life is 31.0 s .

(a) Calculate the concentration of the reactant (in M) 55.8 s after initiation of the reaction.

4.0 ✓

 M

(b) How long (in s) would it take for the concentration of the reactant to drop to one-eighth its initial value?

4.0 ✓

 s

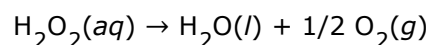
(c) How long (in s) would it take for the concentration of the reactant to drop to 0.0281 M ?

4.0 ✓

 s **Resources**[Read It](#)[Periodic Table](#)

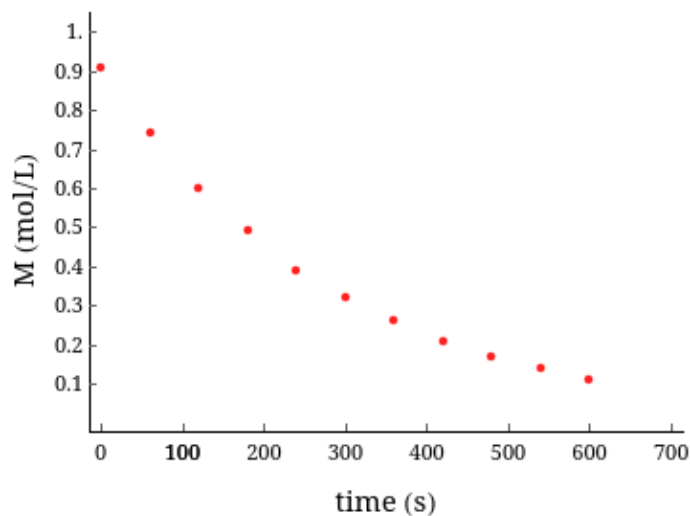
BHChemEng5 11.4.EOC.052.

Hydrogen peroxide (H_2O_2) decomposes into water and oxygen.

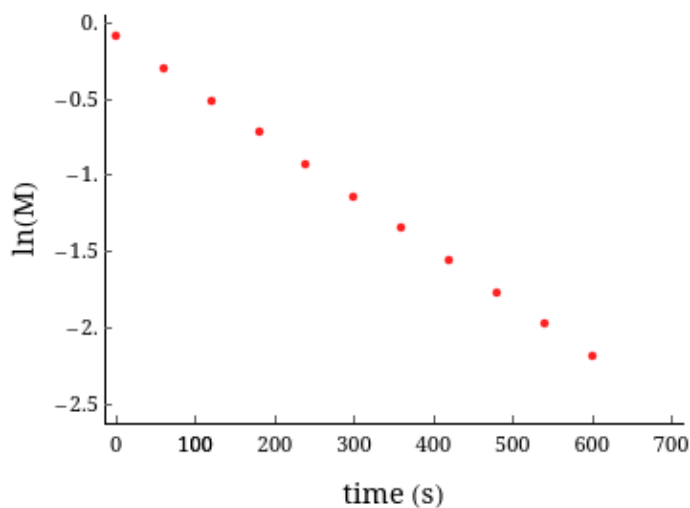


Ordinarily, this reaction proceeds rather slowly, but in the presence of some iodide ions (I^-), the decomposition is much faster.

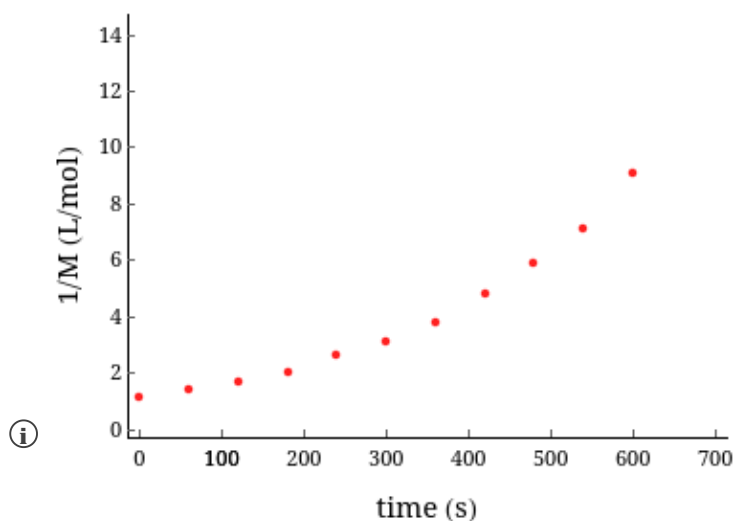
The decomposition in the presence of iodide was studied at 20°C , and the data were plotted in various ways. Use the graphs below to answer the questions that follow.



i



i



(a) What is the order of reaction for the decomposition of hydrogen peroxide?

- ☐ zero-order
☐ first-order
☐ second-order

(b) Find the numerical value of the rate constant at 20°C, including the correct units.

$k =$

- ☐ s^{-1}
☐ $s \cdot L \cdot mol^{-1}$
☐ $mol \cdot L^{-1} \cdot s^{-1}$
☐ $L \cdot mol^{-1} \cdot s^{-1}$

(c) Obtain an estimate of the initial rate of reaction in the experiment that produced the graphs (i.e., the rate at $t = 0$ in the graphs).

rate = $M \cdot s^{-1}$

Resources

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BHChemEng5 11.5.EOC.055.

The following rate constants were obtained in an experiment in which the decomposition of gaseous N_2O_5 was studied as a function of temperature. The products of the decomposition reaction were NO_2 and NO_3 .

k (s^{-1})	Temperature (K)
2.8×10^{-5}	296
1.6×10^{-4}	307
9.0×10^{-4}	319
4.6×10^{-3}	331

Determine E_a for this reaction in kJ/mol.

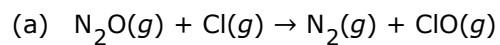
4.0 ✓

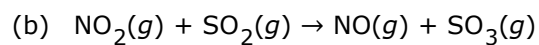
 kJ/mol

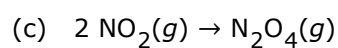
Resources

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What is the rate law for each of the following elementary reactions? (Rate expressions take the general form: $\text{rate} = k \cdot [\text{A}]^a \cdot [\text{B}]^b$.)





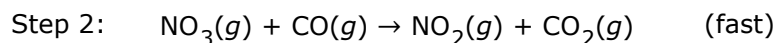
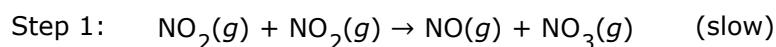


Resources

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The reaction of $\text{NO}_2(g)$ and $\text{CO}(g)$ is thought to occur in two steps.



- (a) What is the overall stoichiometric equation corresponding to this mechanism? (Include states-of-matter under the given conditions in your answer. Use the lowest possible whole number coefficients.)

- (b) What is the molecularity of each step?

(i) Step 1

- ☐ unimolecular
☐ bimolecular
☐ termolecular

(ii) Step 2

- ☐ unimolecular
☐ bimolecular
☐ termolecular

- (c) For this mechanism to be consistent with kinetic data, what must be the experimental rate equation for the overall reaction? (Rate expressions take the general form: $\text{rate} = k \cdot [\text{A}]^a \cdot [\text{B}]^b$.)

- (d) Identify any intermediates in this reaction. (Select all that apply.)

- ☐ CO
☐ CO_2
☐ NO
☐ NO_2
☐ NO_3

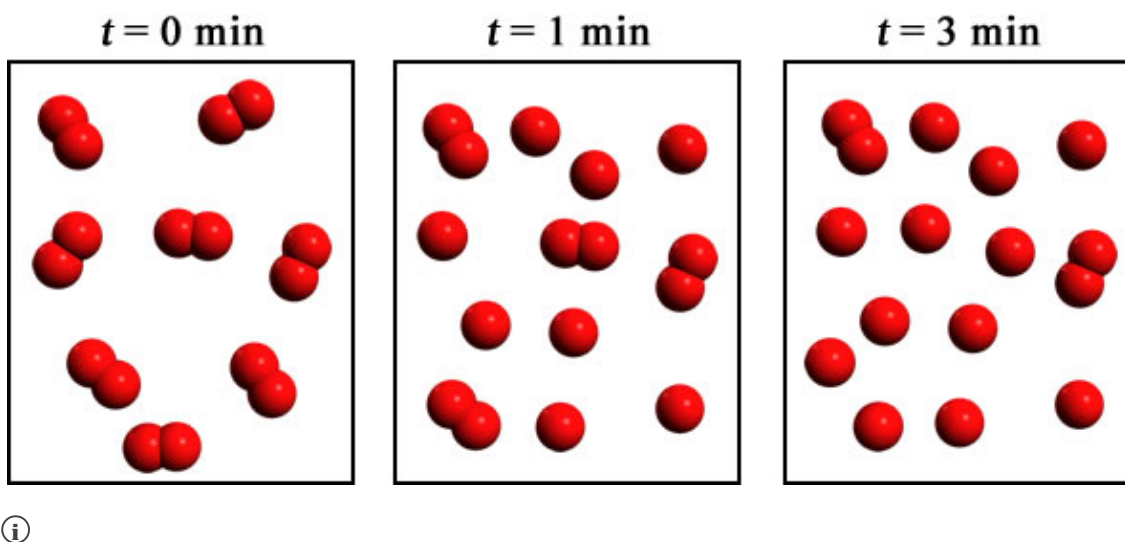
Resources

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BHChemEng5 11.CP.EOC.081.

The following series of pictures represents the progress of a reaction in which A_2 molecules dissociate into atoms: $A_2 \rightarrow 2 A$. Each picture represents a snapshot of the reaction mixture at the indicated time.



Are these figures consistent with a first-order rate law?

- ☐ Yes
- ☐ No
- ☐ Not enough information to say.

Select the most appropriate reasoning for your answer.

- ☐ First-order reactions always occur much faster than the timescale indicated.
- ☐ The half-life is constant.
- ☐ The half-life is not constant.
- ☐ A first-order reaction can only occur between monatomic species.

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